Viren Khandal

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Experience

- **Tesla** — Fremont, CA
  Senior Machine Learning Engineer — *Dec 2022 – Present*
  - Leading ML research and development across 4 global factories, spanning vision-guided robotics, large-scale manufacturing defect detection, and core ML infrastructure for an internal computer vision training platform.
  - Investigating the open problem of factory-aware foundation models – how to learn visual representations that encode manufacturing domain priors and generalize across diverse production environments, inspection modalities, and defect taxonomies.
  - Formulated robust transfer and imitation learning approaches for adapting large pretrained vision models to novel factory domains with minimal labeled data, studying the conditions under which few-shot domain adaptation preserves detection fidelity.
  - Researched knowledge distillation strategies for compressing high-capacity teacher networks into lightweight student models deployable at real-time edge inference latencies, analyzing accuracy–efficiency trade-offs in resource-constrained manufacturing settings.
  - Designed and evaluated production computer vision systems for automated quality inspection, contributing empirical insights on deploying learned models in high-throughput, low-intervention industrial pipelines.
- **Ponder (Stealth)** — Remote
  Co-Founder — *Oct 2024 – Jan 2026*
  - Led full backend and ML development for an AI app that generates multi-perspective audio podcasts from any user query, with each perspective grounded in cited research and credible sources.
  - Architected retrieval-augmented generation pipelines to synthesize web-scale information into structured, multi-viewpoint audio content with source attribution.
  - Built end-to-end backend infrastructure including API design, audio generation, and serving systems to support real-time podcast creation at scale.
- **University of California, Berkeley – VeHICal** — Berkeley, CA
  Research Engineer — *May 2021 – Dec 2022*
  - Led research on safe autonomous-to-human perception handoffs under Professors Sanjit Seshia and Bjoern Hartmann (vehical.org), formalizing stochastic models for verified human-autonomy interaction.
  - Developed VR-based human-in-the-loop experimental tools to study driver takeover scenarios with high ecological validity.
- **Stanford Autonomous Systems Lab** — Stanford, CA
  Visiting Researcher — *Apr 2022 – Dec 2022*
  - Investigated out-of-distribution (OOD) events in human-autonomy interaction under Professor Marco Pavone (stanfordasl.github.io), developing frameworks for safe real-time control handoffs during OOD scenarios.

Education

- **University of California, Berkeley** — Berkeley, CA
  B.A. Computer Science & Applied Mathematics (Double Major) — *Aug 2019 – May 2022*
  - Specialization in Machine Learning and Artificial Intelligence
  - Coursework: Deep Learning, Machine Learning, Artificial Intelligence, Efficient Algorithms, Data Structures, Statistical Prediction, Numerical/Real/Complex Analysis, Advanced Linear Algebra, Abstract Algebra
- **Stanford University** — Palo Alto, CA
  Certificate in Computer Science — *May 2018 – Sept 2018*
  - Coursework: Data Mining and Analysis, Client-Side Internet Technologies

Technical Skills

Languages: Python, C/C++, Go, Java, JavaScript, SQL
ML/AI: PyTorch, TensorFlow, OpenCV, Reinforcement Learning, LLMs, RAG
Infrastructure: AWS, GCP, Docker, Kubernetes